

KAVENNESH BALACHANDAR

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EDUCATION:

Master of Science in Cyber Security | Florida International University, Miami, FL.

August 2024 - April 2026

Bachelor of Computer Science and Engineering | Sri Krishna College of Engineering and Technology, India. **November 2020 – March 2024**

WORK EXPERIENCE:

Student Teaching Assistant | Florida International University, Miami, FL

January 2025 - Present

- Facilitated labs focused on incident response, network security monitoring, and Linux system operations for 60+ students, improving lab completion by 22%.
- Developed training materials and security-focused documentation, increasing course engagement and strengthening technical understanding of threat analysis and access control concepts.
- Provided hands-on troubleshooting support for security configurations, system access issues, and policy enforcement, improving pass rates from 82% to 92%.

Network Security Intern | INTERN DECCAN SOLUTION, Chennai, India

December 2023 - February 2024

- Monitored and optimized Cisco Secure Firewall 3100 across 3 sites, supporting incident response workflows and reducing MTTR by 35%.
- Analyzed security alerts, refined access control policies, and standardized TLS configurations, lowering false positives by 20% and improving risk mitigation.
- Automated 12+ Linux administrative tasks (Bash/Python), improving operational efficiency and accelerating remediation workflows by 25%.
- Developed runbooks and defined monitoring thresholds for security alerts, strengthening documentation, root-cause analysis, and incident escalation processes.

PROJECTS:

Home Networking Lab - Personal Project | Salem, Tamil Nadu, India

May 2025 - July 2025

- Built a Debian-based security lab to simulate network traffic monitoring and access control scenarios, implementing RBAC and secure logging.
- Automated service recovery and monitoring using systemd, improving reliability and reducing configuration overhead by 70%.
- Configured RBAC, secure logging, and health checks to support troubleshooting and system observability.

VPN Monitoring & Detection System | Coimbatore, Tamil Nadu, India

January 2024 - March 2024

- Designed a DPI-based monitoring system to analyze encrypted VPN traffic, identify policy violations, and support incident response across 100+ sessions with 95% accuracy.
- Integrated ELK for real-time log analysis, threat visibility, and alert triage, reducing incident response time from 30 to 8 minutes.
- Implemented structured escalation workflows and mitigation procedures to restore service stability and reduce operational risk.

Enhanced Image Generator | Coimbatore, Tamil Nadu, India

January 2023 - March 2023

- Processed 1,000+ datasets using OpenCV and NumPy, improving classification accuracy through structured preprocessing and performance tuning.
- Automated data processing pipelines to improve operational efficiency and consistency.

PUBLICATION:

VPN Blocker: Detection and Prevention of VPN Tunneling in Enterprise Networks.

February 2024 - March 2024

- Developed a DPI-based threat detection framework to identify encrypted VPN tunnels (90–95% accuracy) and mitigate unauthorized access risks.
- Integrated ELK-based log monitoring and alerting to support threat intelligence analysis and remediation workflows.
- Documented mitigation strategies, root-cause findings, and policy recommendations to strengthen enterprise security posture and reduce MTTR by 30–40%.

CERTIFICATIONS:

- Certified Network Security Practitioner (CNSP) – Merit (**October 2025**)
- Certified Cloud Security Practitioner - AWS (CCSP-AWS) – Merit (**January 2026**)

AWARDS AND EXTRA-CURRICULAR ACTIVITIES:

- Eraksha Cybersecurity Competition (Aug 2023): 1st place nationally for solving real-world cybersecurity challenges.
- Kavach Hackathon (Jun 2023): Qualified at college level for India's leading cybersecurity innovation program.
- Amazon AppSec CTF – NAMER (Sep 2025): Selected to compete globally in Amazon's Application Security Capture the Flag.

SKILLS AND STRENGTHS:

- Programming: Python, Bash, SQL
- Security Tools: Wireshark, Tcpdump, Nmap, Snort, Burp Suite, Metasploit, Nessus
- Network & Protocols: TCP/IP, HTTP, SMTP, FTP, DNS, OpenVPN
- Systems & Automation: Linux (Debian/Kali/Ubuntu), Windows Server/10, systemd, Docker, GitHub Actions
- Security Focus: Network Traffic Analysis, Threat Detection, Malware Analysis, IDS/IPS, VPN Detection, Logging, RBAC